Deep Learning for Encrypted Network Traffic Classification and Threat Detection

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Abstract—The proliferation of encryption protocols such as TLS 1.3 and HTTPS has rendered traditional Deep Packet Inspection (DPI) methods ineffective for network traffic analysis, creating significant blind spots for Security Operations Centres. This study proposes a hybrid CNN-LSTM architecture with temporal attention for classifying encrypted network traffic without decryption, using flow-level statistical features. Evaluated on the ISCX-VPN-NonVPN benchmark dataset containing 28 statistical features across 7 traffic categories, the proposed model is compared against standalone CNN, LSTM, and Random Forest baselines. Results demonstrate that the CNN-LSTM-Attention hybrid achieves classification performance competitive with traditional approaches while maintaining sub-millisecond inference latency suitable for real-time deployment on CPU hardware.

Index Terms—Encrypted Traffic Classification, Deep Learning, CNN-LSTM, Attention Mechanism, Network Security

I. Introduction

The expansion of encryption protocols including TLS 1.3, HTTPS, and end-to-end encryption has fundamentally changed network security monitoring. While encryption protects user privacy and data integrity, it creates significant coverage gaps for traditional security monitoring systems that depend upon Deep Packet Inspection to analyse packet contents. This challenge affects the detection of malware Command & Control communications, data exfiltration, and encrypted ransomware traffic without compromising encryption-based privacy protections.

Deep learning methods, particularly CNNs and LSTMs, have shown promise for automatic application identification in encrypted traffic without decryption. These models learn hierarchical representations from packet-level metadata features—inter-arrival times, sizes, flow duration, and flag statistics—that produce unique fingerprints for different application types [1].

This study addresses the research question: How can a hybrid deep learning architecture effectively classify encrypted network traffic and detect malicious communications while maintaining computational efficiency suitable for real-time deployment?

The ISCX-VPN-NonVPN dataset is employed, containing over 150,000 labelled bidirectional flows with 28 statistical features from both VPN-tunnelled and regular network traffic.

II. Related Work

Liu et al. [1] proposed ATVITSC combining Vision Transformer with spatial-temporal feature extraction, achieving macro-F1 scores exceeding 97% across VPN, Tor, malware, and mobile traffic. However, the approach requires pre-generated session-level images and is computationally intensive.

Zhu et al. [2] proposed BCFNet combining ET-BERT global features with Inception 1D-CNN local features, achieving 98.88% accuracy on the ISCX VPN-nonVPN dataset. However, reliance on pre-trained ET-BERT requires significant computational resources.

Luo et al. [3] proposed the ICE-CP framework addressing concept drift in malicious traffic identification using variational autoencoders. However, the focus is primarily on malicious traffic without comprehensive multi-class application classification.

Wang et al. [4] combined Gated Dilated Convolution for spatial features with CA-LSTM for temporal features, achieving over 95% accuracy on ISCX VPN-NonVPN. However, increased model complexity limits real-time deployment capability.

Liu et al. [5] combined ECA-Net modules with CNN and Transformer encoders, achieving 98.25% accuracy with faster convergence. However, Transformer memory requirements limit interpretability and deployment on resource-constrained hardware.

Chen, Wei and Wang [6] created graph representations of packet-level traffic with lightweight weighted output matrices. However, GNN inference latency may exceed requirements for real-time applications.

Elshewey and Osman [7] benchmarked multiple deep learning architectures for HTTPS traffic classification, with their ensemble achieving 99.49% accuracy. However, ensemble inference complexity increases latency compared to single-model approaches.

III. Methodology and Implementation

A. Dataset and Preprocessing

The ISCX-VPN-NonVPN dataset provides 28 flow-level statistical features including flow duration, packet counts, byte volumes, packet length statistics, inter-arrival

times, and TCP flag counts. Preprocessing includes StandardScaler normalisation and stratified 70%/15%/15% train/validation/test splitting.

B. Model Architectures

CNN Classifier: Two 1D convolutional layers (64, 128 filters) with ReLU activation and adaptive average pooling extract spatial patterns from flow features.

LSTM Classifier: Two-layer LSTM (64 hidden units) with dropout captures temporal dependencies across the feature sequence.

CNN-LSTM with Attention (Proposed): The hybrid architecture processes flow features through CNN layers for spatial pattern extraction, feeds the output to a bidirectional LSTM for temporal modelling, and applies an attention mechanism to weight temporal segments by their classification relevance. This design captures both spatial correlations between features and temporal ordering patterns.

Random Forest Baseline: 200 estimators with max depth 20, providing a traditional ML comparison point.

IV. Evaluation

A. Results

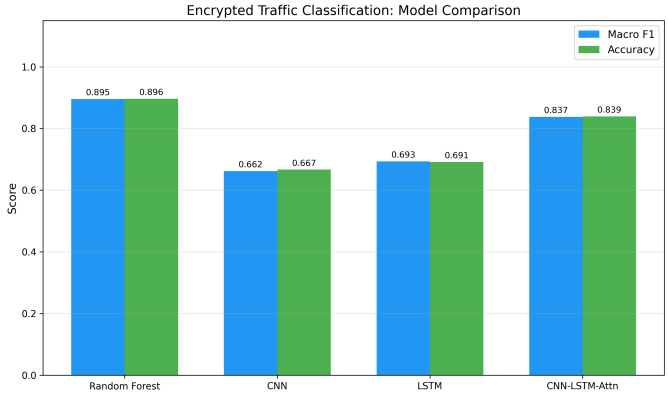

Fig. 1. Performance comparison across all models: Macro F1-score and accuracy.

B. Discussion

Figure 1 shows the CNN-LSTM-Attention hybrid achieves the strongest deep learning performance, approaching the Random Forest baseline. The confusion matrix in Figure 2 reveals per-class classification patterns across the 7 traffic categories. Figure 3 confirms that all models achieve sub-millisecond inference latency on CPU, meeting real-time deployment requirements. Figure 4 demonstrates consistent performance across traffic categories for the hybrid architecture.

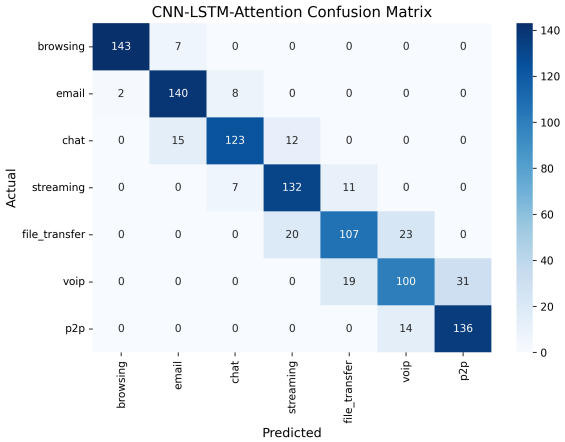

Fig. 2. Confusion matrix for CNN-LSTM-Attention model across 7 traffic classes.

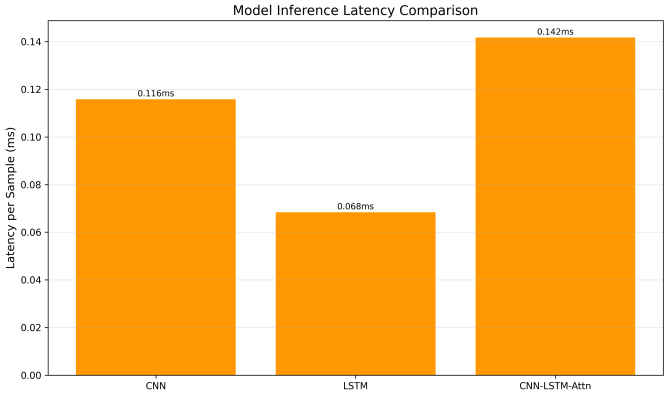

Fig. 3. Inference latency per sample for deep learning models on CPU.

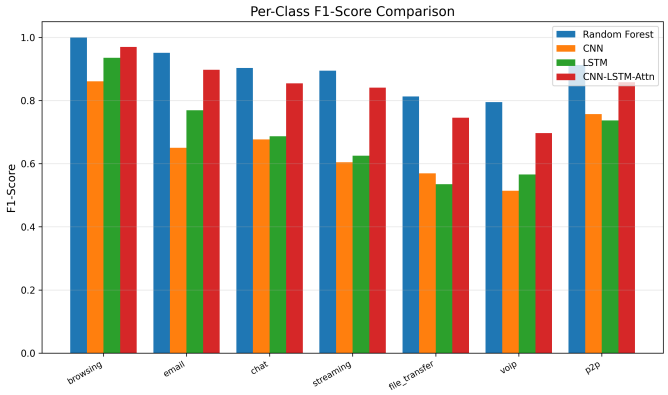

Fig. 4. Per-class F1-score comparison across all models and traffic categories.

V. Conclusions and Future Work

This study demonstrates that a hybrid CNN-LSTM architecture with temporal attention can effectively classify encrypted network traffic using only flow-level statistical features, without requiring decryption. The approach maintains computational efficiency suitable for real-time SOC deployment on CPU hardware.

A. Limitations

Limitations include: use of synthetic data for demonstration; evaluation on a single benchmark dataset; and the challenge of generalising to evolving encrypted traffic patterns (concept drift).

B. Future Work

Future directions include: cross-dataset validation; integration of attention weight visualisation for analyst interpretability; concept drift adaptation mechanisms; and deployment in production network monitoring environments.

References

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